Influenza Outbreak Event Prediction via Twitter

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Introduction

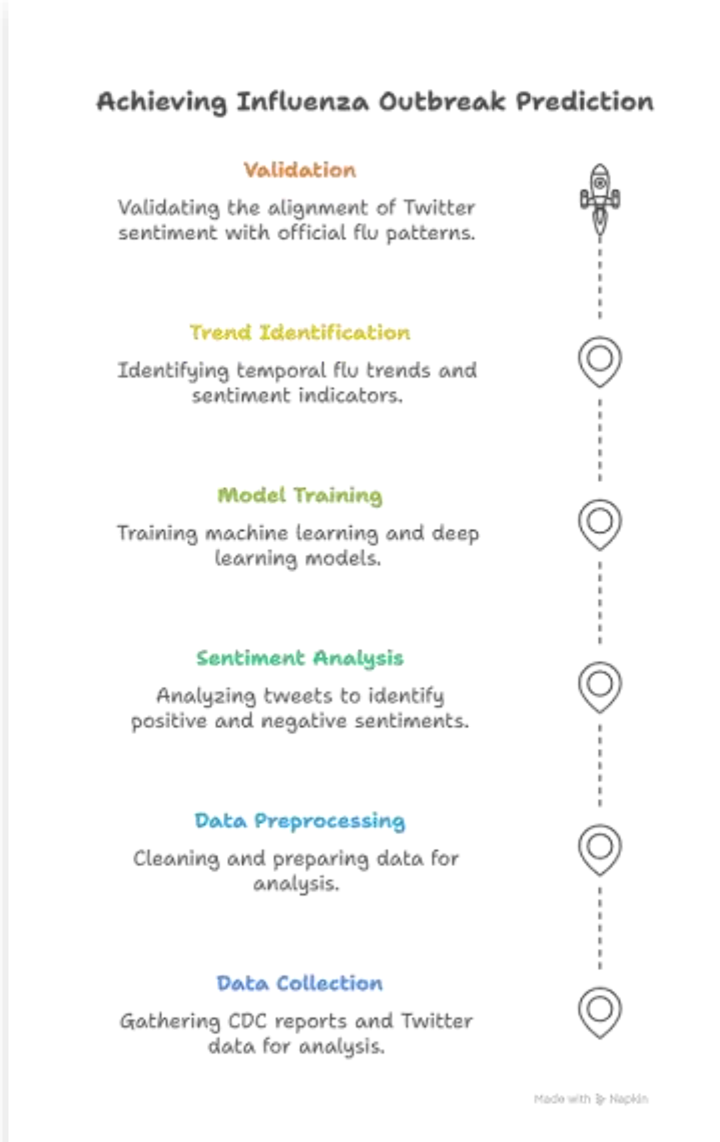
- **Influenza outbreaks** remain a recurring global public health concern.
- **Early prediction** enables faster response and reduces impact.
- **Traditional models** face delays and struggle with real-time, large-scale data.
- This project uses **Twitter data** and **CDC reports** to train ML and DL models.
- Combines **NLP, sentiment analysis (VADER), and LSTM** for outbreak forecasting.

Problem Statement

- Traditional disease surveillance systems rely on delayed clinical/lab data.
- There is no real-time system to detect early influenza outbreak trends.
- Social media platforms like Twitter provide instant public signals, but are noisy and unstructured.
- The challenge: Integrate official CDC data with Twitter-based sentiment analysis to forecast outbreaks early and accurately using ML/DL models.

Objectives

- Forecast influenza outbreaks using CDC reports and Twitter data.
- Perform NLP-based cleaning and VADER sentiment analysis on tweets.
- Train and compare ML (Logistic Regression & Random Forest) and DL models (RNN, LSTM).
- Identify temporal flu trends and key sentiment indicators (e.g., keyword spikes).
- Validate whether Twitter-based sentiment aligns with official flu outbreak patterns.

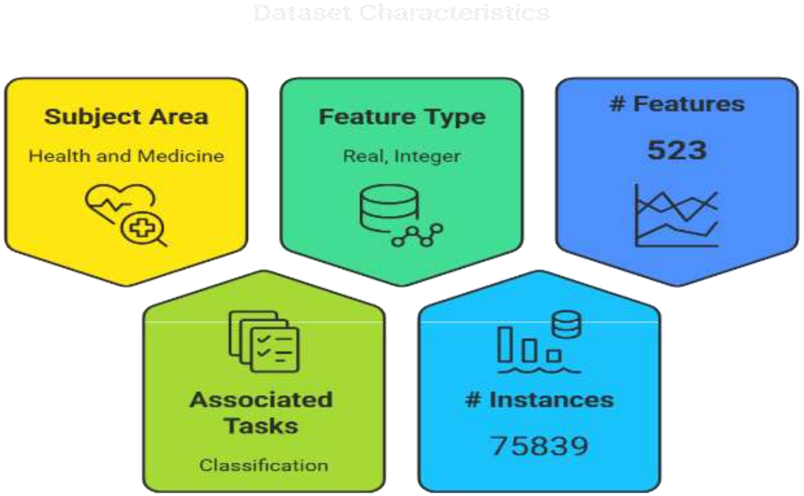


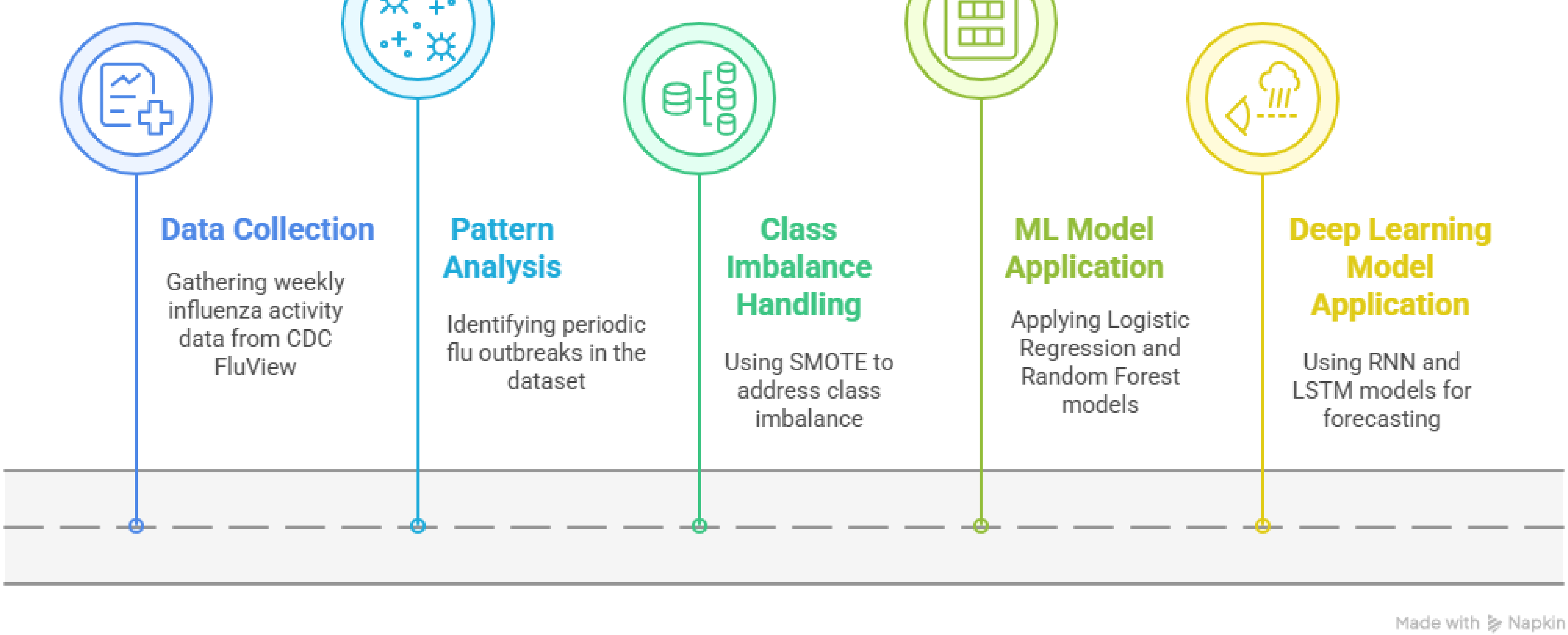
CDC Dataset Overview

The input of the prediction task is the set of the keyword counts for all the tweets in a state in a week. The output is the occurrence of influenza outbreak for the specific state in the next week, which is zero if no event in the next week; or one, otherwise. Here are the briefs of all the variables:

- **'flu_locations':** a list of states.
- **'flu_keywords':** keyword list.
- **'flu_X_*':** input data for all the locations and all the weeks.
- **'flu_Y_*':** output data for all the locations and all the weeks.

525 keywords specified in the variable 'flu_keywords' in the data

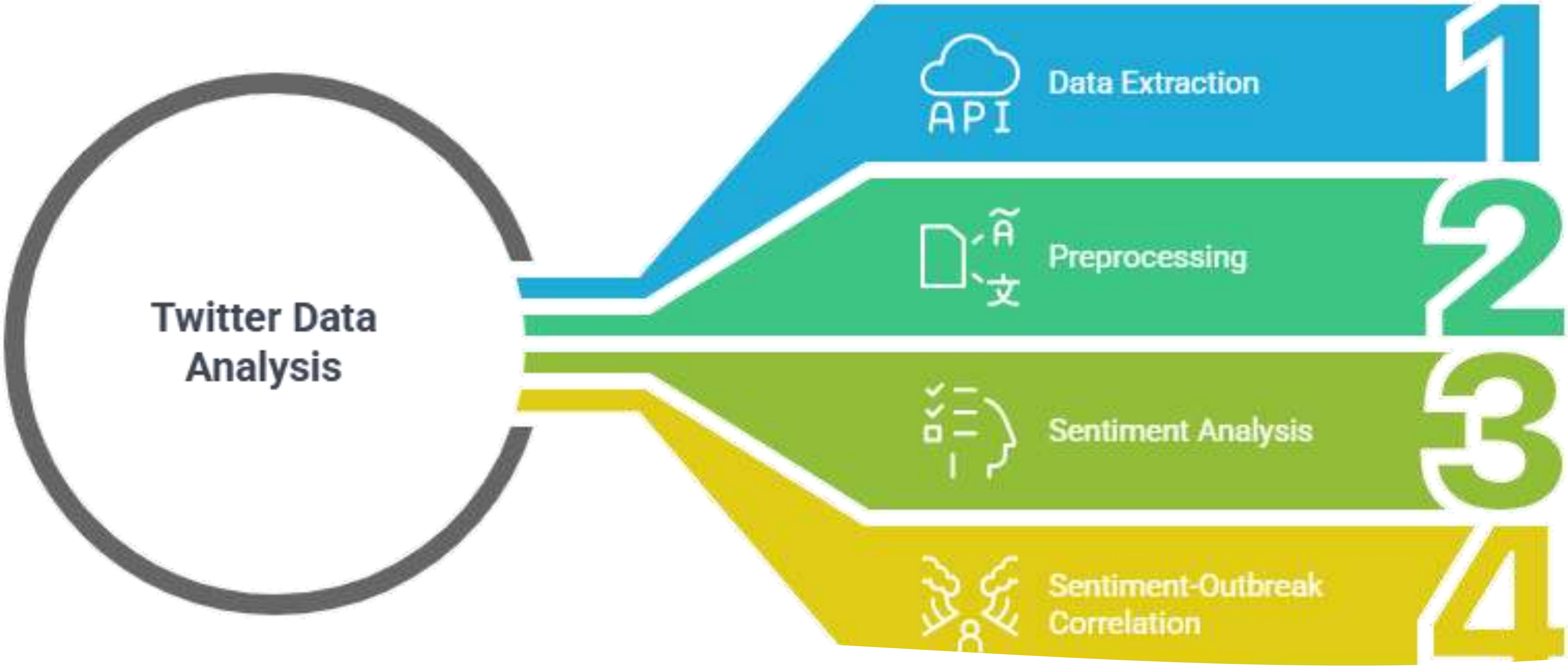




Methodology

Phase 1: CDC Data Modeling

- Weekly influenza activity data collected from CDC FluView.
- Pattern Analysis: Dataset showed periodic flu outbreaks.
- Class Imbalance: Handled using SMOTE.
- ML Models: Logistic Regression, Random Forest.
- Deep Learning Models: RNN & LSTM model for forecasting flu activity.



Methodology

Phase 2: Twitter Data Analysis

- Extracted tweets via Tweepy API (flu, fever, cough).
- Preprocessing: Tokenization, lemmatization, stopword removal.
- Sentiment Analysis using VADER (Positive/Negative).
- Observed that negative sentiment often preceded CDC outbreaks by 1–2 weeks.

Key Features of the Project

Hybrid flu prediction model using CDC + Twitter data.	Deep Learning (LSTM) for time-series outbreak forecasting.	Real-time sentiment tracking via VADER.
Automated tweet preprocessing using NLP.	Regional analysis supported via tweet location metadata.	Early-warning signals detected through negative sentiment spikes.

Conclusion

- Twitter and CDC data combined offer a powerful predictive signal.
- LSTM outperformed traditional models in flu trend forecasting.
- Sentiment trends on Twitter align with official flu outbreaks.

Future Work

- Develop a real-time outbreak dashboard.
- Expand to India/global datasets.
- Apply transformer models (BERT) for better sentiment analysis.
- Extend framework to other diseases like COVID-19 or RSV.

References

- Dataset can be found on Influenza Outbreak Event Prediction via Twitter - UCI Machine Learning Repository
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